

1. A Uniform Resource Locator (URL) is the address of a resource on the Internet.

(a) Explain the different parts of this address:

<https://www.seab.gov.sg/home/examinations/gce-a-level>

(i) <https://>

(ii) [www.](https://)

(iii) [seab.gov.sg](https://)

(iv) [/home/examinations/gce-a-level](https://)

(b) Explain how the DNS is used when this URL is entered into a web browser.

(a) (1.2)

<code>http://</code>	Protocol for accessing resource (method)
<code>www</code>	Server on which resource is stored (hostname)
<code>seab.gov.sg</code>	Domain name
<code>/home/examinations/gce...</code>	Path of resource (it's fine that no filename is specified, the server is configured to know what HTML file to serve to the client)

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(i) <https://>

(ii) [www.](https://www)

(iii) [seab.gov.sg](https://www.seab.gov.sg)

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(b) Explain how the DNS is used when this URL is entered into a web browser.

(b) (1.5)

- DNS is like a phonebook: look up a domain name and it converts to an IP address
- Check cache local DNS server (at ISP) → can find, return to user the IP address for user device to go to webserver.
- If cannot find, recursively go to next level larger DNS database up the hierarchy to resolve domain name (regional ISP, network service providers)
- Once found, then return IP address to local DNS server (thru the higher levels DNS server) which caches this IP info for future references

2. A voluntary organisation is considering purchasing a lease on a web domain to set up a new website to advertise their events and services. They have been advised to contact an Internet registrar. Explain the role of an Internet registrar.

(1.3)

- provide domain name registration services to individuals, businesses, and organisations
- act as intermediaries between customers and the domain name registries, facilitating the process of acquiring, managing, and renewing domain names.

3. All Internet connected devices communicate via the TCP/IP protocol stack. This has four layers.
- (a) Describe the roles of each layer when two devices are communicating over the Internet.
 - (b) What are the hardware that operate on the (i) Network and (ii) Data link layer?

(a) (5.1 – 5.4)

The **Application layer** selects the appropriate protocol (e.g., HTTP) for data transmission and formats the text data for communication between the application and the network, ensuring compatibility and readability.

The **Transport layer** employs TCP for reliable end-to-end connections, by segmenting data into packets labeled with sequence numbers to ensure ordered delivery, and port numbers for routing to the appropriate application on the receiving device. It utilises acknowledgements to verify success packet reception and trigger retransmission if necessary.

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The **Network layer** assigns source and destination **IP addresses** to packets. Routers utilise these addresses to forward packets to their destination. Combining an IP address with a port number creates a socket, such as "113.10.64.21:80," specifying the device and web browser application running on a device with IP address 113.10.64.21.

The **Data Link layer** assigns **MAC addresses** to devices, enabling hardware identification for packet delivery, and handles physical connections between network nodes, facilitating data transmission.

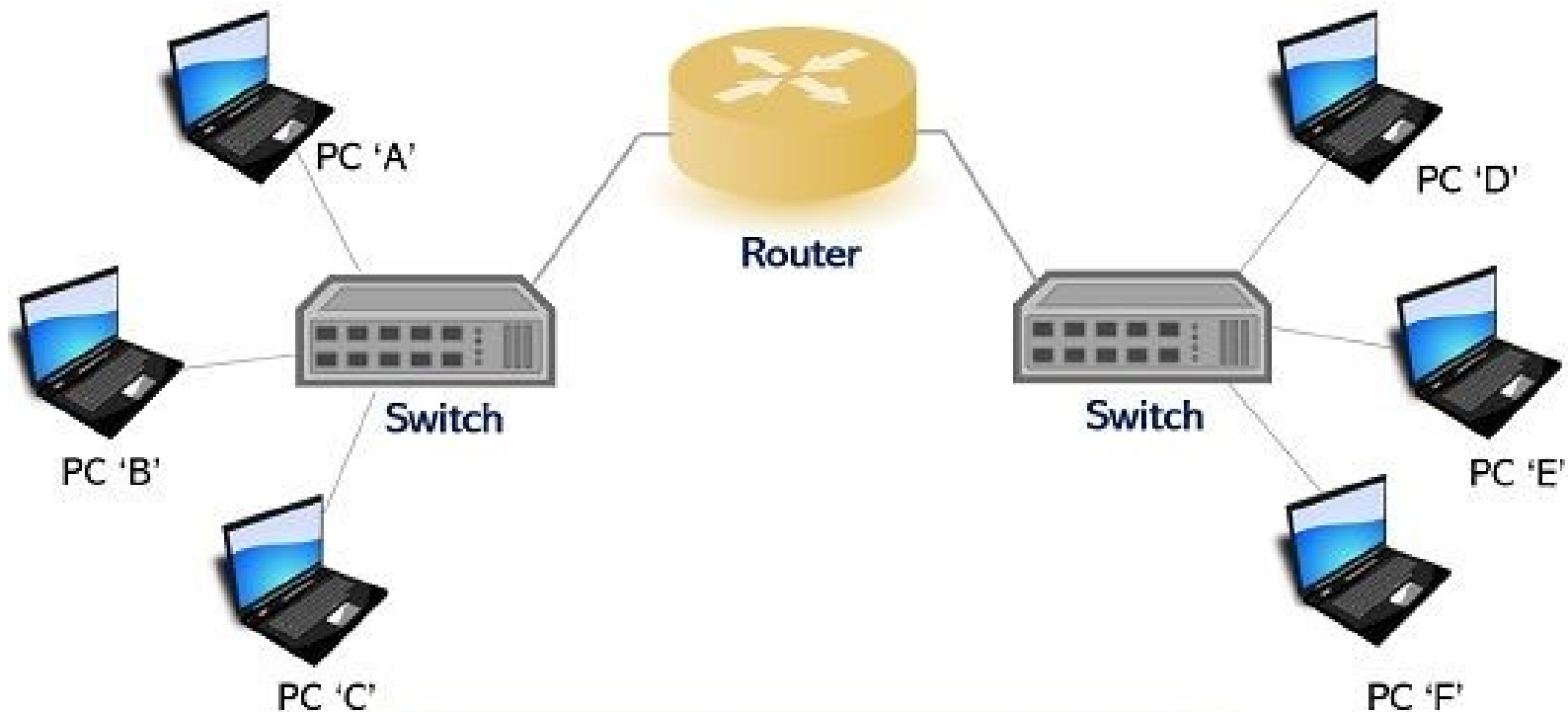
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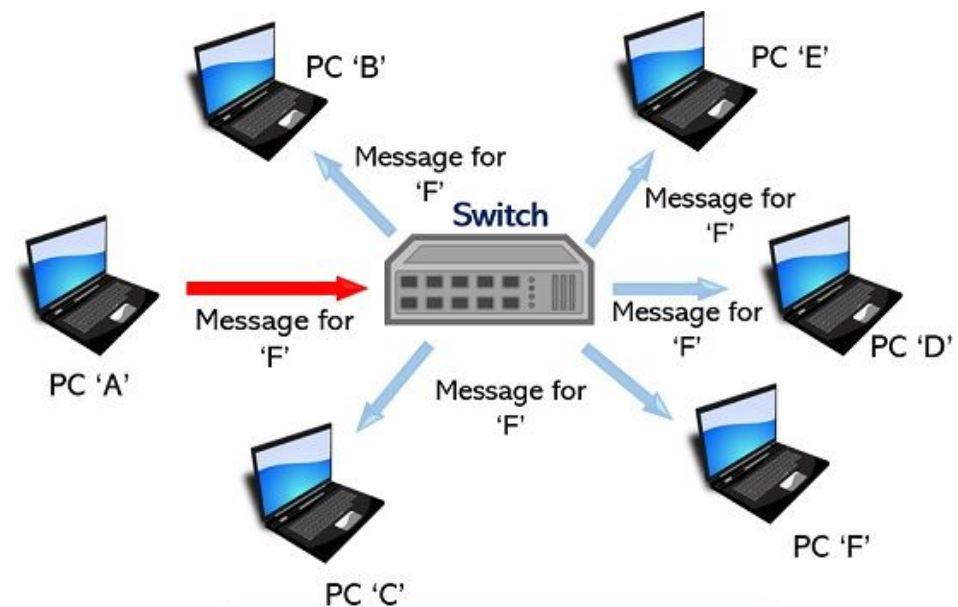
(i) Router

(ii) Switch

A router connects different networks together, facilitating data exchange between them, while a switch connects multiple devices within a single network, enabling communication among them.



Connection of networks through Router



Data reaching connected devices through switch

4. Major parts of the Internet run on a packet switched network that relies on routers and gateways to communicate.
- (a) What is the difference between a router and a gateway?
 - (b) What is meant by the term packet switching?
 - (c) A data packet contains a header and a payload. The header contains data that it used to route the packet to its destination. State three data items that might be contained in a data packet's header.

(a) (4.6 and 4.7)

Router: route a data packet across a network to its correct destination

Gateway: similar function as router, but connects two networks with different transmission protocols

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(b) (4.4) Packet switching

Transmission in small data packets

Each packet takes best route available at any given time, not established route

Packets may not arrive in sequence at destination, need to reassemble at destination

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(c) (4.5)

- Sender's and destination's IP address
- Protocol being used, e.g TCP
- Packet number in sequence (not total number of packets)
- Time To Live or hop limit

5. Explain the difference between client-server and peer-to-peer networking, and give an example of where each might be used.

(6.1), (8)

Client-server: Has a centralised server / Peer-to-peer: No central server

Many computers connect to a powerful central computer known as server.

Each client holds some of its own files and software, and also access resources from server.

Server stores, manages, and delivers these shared resources to the clients.

Dedicated resources and centralised control are necessary.

Web server, file server, cloud computing services.

No central server, individual computers connected to each other through a network.

Computers share files directly, no central computer manages resources.

File sharing, real-time collaboration, distributed resource sharing are desired.

Content distribution such as video on-demand.

6. Explain three advantages of using cloud-based services?

(6.2)

- **Scalability:** Resources can be increased or decreased as needed, and users pay only for what they use.
- **Accessibility:** Data and applications can be accessed from any device and location.
- **Cost Efficiency:** Eliminates the need for expensive hardware and reduces maintenance costs.

7. A supermarket is designing a website which will allow its customers to place orders online. Customers will login to the website using email address and it will be validated when an order is made.
 - (a) The individual web pages that describe each product will be generated dynamically using server-side scripting. What is a server-side scripting?
 - (b) Client and server side validation should both be done. Explain why it is important to validate the email address again once it reaches the server.

(a) (6.3)

Web pages generated on server side and send over to client.

Server has all the resources and processing power to interact with database, perform calculations, generate web pages based on client's request and needs, and send to client for minimal processing.

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(b) (6.3)

Data integrity and security. In case client-side validation is circumvented maliciously, the server-side needs to validate again. Sometimes, client-side validation may be limited due to the client's devices.

Client-side validation is usually very basic check, for example on format of email address. When sent to server, there may be greater security check, for example, on the effect that the email address may have on the SQL query that may be generated, in case it becomes malicious line of SQL code.

8. Explain two differences between a LAN and a WAN.

- (section 2)
- Computers are relatively closely situated **vs** geographically dispersed
- Connect using wireless links, wire cables, fibre optic cables **vs** telephone links, undersea cables and satellites
- Faster data transfer, high reliability, lower setup costs, **vs** higher costs, slower speed due to public connections

9. A company has offices in three countries connected through a WAN. Employees in each office access shared files using an internal website.

- (a) Explain why a WAN is required rather than a LAN.
- (b) Explain how routers help data travel between offices.
- (c) Explain why an Intranet would be useful in this situation.

- (a) LAN is small geographic area, like single building or campus. WAN can bridge long distances over countries.
- (b) Router reads the destination IP address of the packet, determine the most efficient path through the global network, hands the packet to the next router until it reaches.
- (c) (section 3) Intranet provides secure, private website accessible to the company's employees.

10. Explain the difference between a MAC address and an IP address.

- (4.8)
- MAC: identifies hardware, IP address: logical location on network
- Scope: local (inside home/office network) vs global (across different network)
- Layer 2 (data link) vs layer 3 (Network layer)

11. Compare static and dynamic IP addresses, including how they are assigned and one advantage and disadvantage of each.

- (1.6)
- Permanent vs temporary
- Manual by network admin vs automatic assigned by DHCP
- Higher cost vs cost effective

12. Explain the difference between thin client and thick client computing.

- (6.4)
- Amount of processing and storage that happens on client devices versus server.
- Thin means lesser on client

13. Explain two differences between an Intranet and the Internet.

- (3)(7.1)

14. Explain how a Virtual Private Network (VPN) allows secure remote access to an organisation's Intranet.

- (3)
- by establishing a **secure, encrypted tunnel** over the public internet between a user's device and the organisation's private network

15. Explain how subnetting reduces IP address wastage in a network.

- (CIDR and subnetting page 7)
- by dividing a large, rigid block of IP addresses into smaller, customized segments that match the actual size of each sub-network

16. Explain the structure of IPv4 and IPv6 addresses, including format, size, and representation.

- (1.4)
- 32 bits **vs** 128 bits
- Four decimal numbers separated by dots **vs** Eight groups of four hexadecimal digits separated by colons
- Dotted decimal notation **vs** Hexadecimal
- Example: 192.168.1.1 vs 2001:0db8:85a3:0000:0000:0000:0370:7334
(2001:0db8:85a3::0370:7334)

17. Explain the function of a router, switch, modem, and gateway in a network.

- (7.2)